

Equivalent to some code examples in C (with some C++ facilities):

<u>For pseudo code example</u>	<u>Equivalent in C language (with some C++ facilities)</u>
Algo Exmpl Var N, i : Integer Begin Read (N) For i = 1 to 5 do N ← N * 2 EndFor Write(N) End	<pre>#include <iostream> using namespace std; int main() { int N, i; cin >> N; for (i = 1; i<=5 ; i++) { N = N * 2; } cout << N; }</pre>
For i = 1 to 5 step 2 do N ← N * 2 EndFor	<pre>for (i = 1; i<=5 ; i+=2) { N = N * 2; }</pre>

<u>For pseudo code example</u>	<u>Equivalent in C language (with some C++ facilities)</u>
For i = 10 to 1 step -3 do N ← N * 2 EndFor	<pre>for (i = 10; i >= 1 ; i+=-3) { N = N * 2; }</pre>
For i = 1 to 3 do Read(N) For j = 1 to 5 do N ← N * 2 EndFor Write(N) EndFor	<pre>for (i = 1; i<=3 ; i++) { cin >> N; for (j = 1; j<=5 ; j++) { N = N * 2; } cout << N; }</pre>

<u>While pseudo code example</u>	<u>Equivalent in C language (with some C++ facilities)</u>
While N < 100 do N ← N * 2 EndWhile	<pre>while (N < 100) { N = N * 2; }</pre>

<u>Repeat pseudo code example</u>	<u>Equivalent in C language (with some C++ facilities)</u>
Repeat N ← N * 2 Until N ≥ 100	<pre>do{ N = N * 2; }while (N < 100);</pre>

Note:

- There are no semicolon after **for** and **while** condition : **for(i=1;i<=5;i++) ; while(N < 100) ;**

```

{
    ...
}

```

- There are a semicolon after **do** condition :

```

do{
    ...
}while (N < 100) ;

```

Do not consider possible user error cases in this series

Solution of Exercise 3.01:

a) <pre> Begin Read(N) S ← 0 For i ← 1 to N do S ← S + i EndFor Write(S) End </pre> For N = 2, S = 3 For N = 3, S = 6 For N = 5, S = 15 For N = 1, S = 1 For N = 0, S = 0 - What does this pseudo code do?	b) <pre> Begin Read(N) F ← 1 For i ← 1 to N do F ← F * i EndFor Write(F) End </pre> For N = 2, F = 2 For N = 3, F = 6 For N = 5, F = 120 For N = 1, F = 1 For N = 0, F = 1 - What does this pseudo code do?
c) <pre> Begin Read(B,P) Res ← 1 For i ← 1 to P do Res ← Res * B EndFor Write(Res) End </pre> For B=3 and P=2, Res= 9 For B=5 and P=2, Res= 25 For B=2 and P=3, Res= 8 For B=0 and P=2, Res= 0 - What does this pseudo code do?	d) <pre> Begin Co ← 0 For i ← 1 to 3 do Read(N) If N mod 2=0 then Co ← Co + 1 EndIf EndFor Write(Co) End </pre> For N=10 then 2 then 3, Co = 2 For N=1 then 5 then 6, Co = 1 For N=9 then 7 then 13, Co = 0 - What does this pseudo code do ?

Solution of Exercise 3.04_b2:

Algo Exo3_4_b2 Var B, Res : Real P, i: Integer Begin <pre> Read(B,P) If P < 0 then P ← -P B ← 1/B EndIf Res ← 1 For i ← 1 to P do Res ← Res * B EndFor Write(Res) End </pre>	For B=2 and P=3, Res= 8 For B=7.5 and P=3, Res= 421.875 For B=2 and P=-3, Res= 0.125 For B=0 and P=5, Res= 0 For B=2.6 and P=0, Res= 1
--	--

Solution of Exercise 3.05_c2:

Algo Exo3_5_c2

Var

x, Res, S : Real

i, F : Integer

Begin

Read(x)

S ← 1

F ← 1

Res ← 1

For i ← 1 **to** 9 **do**

F ← F * i

Res ← Res * x

S ← S + Res/F

EndFor

Write(S)

End

x	S
0	1
1	2.71828
2	7.38871
2.3	9.97275

Exercise:

1. Write a program that displays two lines of stars (by asking for the number of stars).

```
* * * * *
* * * * *
```

2.1 Write a program that displays a square (by asking for the number of stars on one side).

2.2 Write a program that displays a rectangle (by asking for the number of stars of the length and width side).

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

3. Write a program that displays a triangle (by asking for the number of lines)

```
*
**
***
****
*****
```

```
*****
****
***
**
*
```

```
*
**
***
****
*****
```

```
*****
*****
*****
*****
*****
```

```
*****
*****
*****
*****
*****
```

4. Write a program that displays an empty rectangle (by asking for the number of stars of the length and width side)

```
* * * * *
*
*
*
*
* * * * *
```